

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	03 October 2026
Team ID	SWTID-2026-5622
Project Name	Script-Controlled ACL –Restricted Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	ACL Configuration	Create ACL rules for specific tables and operations. Configure read, write, create, and delete permissions.
FR-2	Field-Value Based Access Control	Define conditions based on field values. Allow access when the specified field condition is satisfied. Deny access when the condition is not satisfied.
FR-3	Script Controlled Authorization	Execute authorization scripts during access requests. Validate user roles and field values dynamically. Return Allow/Deny decisions based on script conditions.
FR-4	Access Validation and Logging	Validate access before performing the requested operation. Display an appropriate access-denied message. Record access decisions for auditing and troubleshooting.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	ACL configuration and field-value conditions should be simple to understand and manage. Access-denied messages should clearly indicate that the user does not have the required permission.
NFR-2	Security	The system must enforce authorization before allowing users to access or modify protected records. Unauthorized users must not be able to bypass script-controlled ACL rules.
NFR-3	Reliability	ACL rules and authorization scripts should consistently produce the correct access decision. Failures or invalid conditions should not unintentionally grant unauthorized access.
NFR-4	Performance	Authorization checks should execute with minimal delay and should not significantly affect normal record operations.
NFR-5	Availability	Access-control functionality should remain available whenever users perform operations on protected records.
NFR-6	Scalability	The solution should support increasing numbers of users, records, ACL rules, and field-based conditions without significant degradation in performance.